Quizrr Chapter-Wise Test for JEE Main - 2025

Q.1

Reaction of zinc with cold and very dilute nitric acid yields

(A) $\mathrm{Zn}\left(\mathrm{NO}_3\right)_2+\mathrm{N}_2\mathrm{O}$

(B) $\mathrm{Zn}\left(\mathrm{NO}_3\right)_2+\mathrm{NO}$

(C) $\mathrm{Zn}\left(\mathrm{NO}_3\right)_2+\mathrm{NH}_4\mathrm{NO}_3$

(D) $\mathrm{Zn}\left(\mathrm{NO}_3\right)_2+\mathrm{NO}_2$

Q.2

A copper salt is isomorphic with ZnSO_4, the salt will be

(A) Paramagnetic

(B) Diamagnetic

(C) Ferromagnetic

(D) None

Q.3

Light green crystals of ferrous sulphate lose water molecules and turn brown on exposure to air. This is due to its oxidation to

(A) $\mathrm{Fe}_2\mathrm{O}_3$

(B) $\mathrm{Fe}_2\mathrm{O}_3 \cdot \mathrm{H}_2\mathrm{O}$

(C) $\mathrm{Fe(OH)SO}_4$

(D) $\mathrm{Fe}_2\mathrm{O}_3+\mathrm{FeO}$

Q.4

In which of the following ions, $\mathrm{d}-\mathrm{d}$ transition is not possible ?

(A) Ti^{4+}

(B) Cr^{3+}

(C) Mn^{2+}

(D) Cu^{2+}

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Q.5

The aqueous solution containing which one of the following ions will be colourless

(A) Sc^{3+}

(B) Fe^{2+}

(C) Ti^{3+}

(D) Mn^{2+}

Q.6

At $pH = 4, Cr_2O_7^{2-}$ exists as

(A) CrO_4^{2-}

(B) CrO_3

(C) CrO_2^{2+}

(D) $Cr_2O_7^{2-}$

Q.7

$[Fe\,(H_2O)_5\,NO]^{2+}$ is brown-ring complex in NO_3^-. In this complex:

(A) NO transfers its electron to Fe^{2+} so that
there are three electrons unpaired making iron as $Fe(I)$ and nitrosyl as NO^+

(B) magnetic moment of Fe is $3.87\ BM$

(C) the colour is due to charge transfer

(D) all are correct statements

Q.8

If H_2S gas is passed into a solution of Cu^{2+}, Cd^{2+} having excess of KCN.

(A) CuS and CdS both are precipitated.

(B) soluble complex $[Cu(CN)_4]^{3-}$ and $[Cd(CN)_4]^{2-}$ are formed and no effect of passing H_2S gas

(C) soluble complex $[Cu(CN)_4]^{3-}$ and $[Cd(CN)_4]^{2-}$ are formed, of which CdS is precipitated as yellow ppt.

(D) soluble complex $[Cu(CN)_4]^{3-}$ and $[Cd(CN)_4]^{2-}$ are formed of which CuS is precipitated as black ppt.

Q.9

The catalyst used in the Haber process for the manufacture of ammonia is

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(A) Pt

(B) Pd

(C) Ni

(D) Fe

Q.10

Which of the following does not represent the correct sequence of indicated properties?

(A) $\mathrm{Sc}^{3+} > \mathrm{Fe}^{3+} > \mathrm{Mn}^{3+}$; stability of $+3$ oxidation state

(B) $\mathrm{Mn}^{2+} < \mathrm{Ni}^{2+} < \mathrm{Co}^{2+} < \mathrm{Fe}^{2+}$; magnetic moment

(C) $\mathrm{FeO} > \mathrm{CoO} > \mathrm{NiO};$, basic character

(D) $\mathrm{Se} < \mathrm{Ti} < \mathrm{Cr} < \mathrm{Mn};$ number of oxidation states

Q.11

The Irving-Williams order regarding the stability of the complexes of bivalent metal of the first transition series is

(A) $\mathrm{Mn} < \mathrm{Fe} < \mathrm{Co} < \mathrm{Ni} < \mathrm{Cu} < \mathrm{Zn}$

(B) $\mathrm{Mn} > \mathrm{Fe} > \mathrm{Co} > \mathrm{Ni} > \mathrm{Cu} > \mathrm{Zn}$

(C) $\mathrm{Mn} < \mathrm{Fe} < \mathrm{Co} > \mathrm{Ni} > \mathrm{Cu} > \mathrm{Zn}$

(D) $\mathrm{Mn} > \mathrm{Fe} > \mathrm{Co} < \mathrm{Ni} < \mathrm{Cu} < \mathrm{Zn}$

Q.12

Which of the following ions is expected to be diamagnetic ?

(A) ${}_{25}\mathrm{Mn}^{2+}$

(B) ${}_{29}\mathrm{Cu}^{+}$

(C) ${}_{27}\mathrm{Co}^{3+}$

(D) ${}_{24}\mathrm{Cr}^{3+}$

Q.13

Which of the following statements about the complex $\left[\mathrm{CoF}_6\right]^{3-}$ which is paramagnetic in nature is correct?

(A) Cobalt involves $\mathrm{d}^2\mathrm{sp}^3$ hybridization

(B) Cobalt involves $\mathrm{sp}^3\ \mathrm{d}^2$ hybridization

(C) F is in zero oxidation state

(D) Co is in $+2$ oxidation state

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Q.14

A transition element containing only one electron in $4s$ orbital is

(A) $_{21}Sc$

(B) $_{23}Mn$

(C) $_{26}Fe$

(D) $_{29}Cu$

Q.15

The compound which have the same magnetic moment like that of $FeCl_2$ is

(A) $CrCl_3$

(B) $MnCl_2$

(C) $CoCl_3$

(D) $NiCl_2$

Q.16

Which of the following compounds is likely to be coloured ?

(A) $TiCl_3$

(B) $Ti\,(SO_4)_2$

(C) Cu_2Cl_2

(D) $ScCl_3$

Q.17

In $MnO_4^- + I^- \xrightarrow{\text{Alkmed.}} X; X$ is:

(A) IO_3^-

(B) IO_4^-

(C) I_2

(D) IO^-

Q.18

Which of the following is diamagnetic in nature ?

(A) $CrCl_3$

(B) $ZnCl_2$

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(C) $NiCl_4^{2-}$

(D) $CuSO_4$

Q.19

The basic nature of the monoxides of the first transition series is of the following order

(A) $CrO > VO > FeO > TiO$

(B) $TiO > VO > CrO > FeO$

(C) $VO > CrO > TiO > FeO$

(D) $TiO > FeO > CrO > VO$

Q.20

Maximum magnetic moment is shown by:

(A) d^5

(B) d^6

(C) d^7

(D) d^8

Q.21

Zinc reacts with excess of caustic soda to form

(A) $Zn(OH)_2$

(B) ZnO

(C) Na_2ZnO_2

(D) $Zn(OH)_2ZnCO_3$

Q.22

The most of inner-transition elements, show the valency of

(A) $+2$

(B) $+3$

(C) $+4$

(D) $+5$

Q.23

Arrange $Ce^{+3}, La^{+3}, Pm^{+3}$ and Yb^{+3} in increasing order of their ionic radii

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(A) $Yb^{+3} < Pm^{+3} < Ce^{+3} < La^{+3}$

(B) $Ce^{+3} < Yb^{+3} < Pm^{+3} < La^{+3}$

(C) $Yb^{+3} < Pm^{+3} < La^{+3} < Ce^{+3}$

(D) $Pm^{+3} < La^{+3} < Ce^{+3} < Yb^{+3}$

Q.24

Among the following series of transition metal ions, the one in which all metals ions have $3d^2$ electronic configuration is

(A) $Ti^{4+}, V^{3+}, Cr^{2+}, Mn^{3+}$

(B) $Ti^{2+}, V^{3+}, Cr^{4+}, Mn^{5+}$

(C) $Ti^{3+}, V^{2+}, Cr^{3+}, Mn^{4+}$

(D) $Ti^{+}, V^{4+}, Cr^{6+}, Mn^{7+}$

Q.25

Transuranic elements begin with

(A) Np

(B) Cm

(C) Pu

(D) U

Q.26

Due to lanthanide contraction following pair(s) may have approximately equal size:

(A) Zr and Hf

(B) Nb and Ta

(C) Mo and W

(D) all of these

Q.27

Which group of metals is called uranides:

(A) Fe, Co, Ni

(B) Ce, Pr, Nd

(C) Ac, Th, Pa

(D) Pu, Am, Cm

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Q.28

Across lanthanide series the basicity of lanthanide hydroxides

(A) Decreases

(B) Increases

(C) First decreases and then increases

(D) First increases and then decreases

Q.29

The total number of rare earth elements is :

(A) 8

(B) 14

(C) 32

(D) 10

Q.30

An element has electronic configuration $1s^2\ 2s^2\ 2p^6\ 3s^2\ 3p^6\ 3d^{10}\ 4s^2\ 4p^6 4d^{10}\ 5s^2\ 5p^6\ 5d^1\ 4f^7\ 6s^2$. In which group should it be placed ?

(A) Fifth

(B) Thirteenth

(C) Third

(D) Seventeenth

Answers & Solutions

Q.1 Answer:

$Zn\left(NO_3\right)_2 + NH_4NO_3$

Solution:

Very dil HNO_3 $:\rightarrow$ **Ammonium nitrate is formed**

$4Zn + 10HNO_3 \longrightarrow 4Zn\left(NO_3\right)_2 + NH_4NO_3 + H_2O$

Q.2 Answer:

Paramagnetic

Solution:

$ZnSO_4 \longrightarrow Zn^{++} + SO_4^{-2}$

$Zn^{++} \approx Cu^{++}$

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$Cu^{+2} \longrightarrow 3d^9 - 1$ **unpaired** e^-

$\therefore$ paramagnetic in nature.

Q.3 Answer:

$Fe(OH)SO_4$

Solution:

$Fe(OH)SO_4$

$FeSO_4 \xrightarrow{H_2O} Fe(OH)SO_4$

Q.4 Answer:

Ti^{4+}

Solution:

Due to absence of d-electron in Ti^{4+} $\left([Ar]^{18}3d^04s^0\right)$, $d - d$ **transition is not possible.**

Q.5 Answer:

Sc^{3+}

Solution:

$_{21}Sc \rightarrow 1s^2\ 2s^2 2p^6 3s^2 3p^6 4s^2 3d^1$

$Sc^{+3} \rightarrow 1s^2 2s^2 2p^6 3s^2 3p^6$

Due to the absence of unpaired electrons it will be colourless.

Q.6 Answer:

$Cr_2O_7^{2-}$

Solution:

$pH = 4$ **(i.e., acidic medium) no change but**

at $pH = 12$ **(i.e., basic medium)**

$Cr_2O_7^{2-}$ **changes to** CrO_4^{2-}

$Cr_2O_7^{2-} + 2OH^- \longrightarrow 2CrO_4^{2-} + H_2O$

Q.7 Answer:

all are correct statements

Solution:

$NO \longrightarrow NO^+ + e^-$

$\underset{3d^6}{Fe^{2+}} + e^- \longrightarrow \underset{3d^7}{Fe^+}$

four unpaired electron — three unpaired electrons

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Magnetic moment of $\mathrm{Fe}^{+} = \sqrt{15}\ \mathrm{BM}$
$= 3.87\ \mathrm{BM}$

Q.8 Answer:

soluble complex $[\mathrm{Cu(CN)_4}]^{3-}$ and $[\mathrm{Cd(CN)_4}]^{2-}$ are formed, of which CdS is precipitated as yellow ppt.

Solution:

soluble complex $[\mathrm{Cu(CN)_4}]^{3-}$ **and** $[\mathrm{Cd(CN)_4}]^{2-}$ **are formed, of which** CdS **is precipitated as yellow ppt.**

Q.9 Answer:

Fe

Solution:

Fe **is used as a catalyst in the Haber process.**

Q.10 Answer:

$\mathrm{Mn^{2+}} < \mathrm{Ni^{2+}} < \mathrm{Co^{2+}} < \mathrm{Fe^{2+}}$; magnetic moment

Solution:

The number of unpaired electrons of the ions in the given choices are $\mathrm{Mn^{2+}(n=5)}; \mathrm{Ni^{2+}(n=2)}$ **,** $\mathrm{Co^{2+}(n=3)}$ **and** $\mathrm{Fe^{2+}(n=4)}$**.**
The correct sequence of magnetic moments would be $\mathrm{Mn^{2+} > Fe^{2+} > Co^{2+} > Ni^{2+}}$**.**

Q.11 Answer:

$\mathrm{Mn < Fe < Co < Ni < Cu < Zn}$

Solution:

For a given ligand the stability of the complexes of the metallic ions having the same charge decreases with the increase in the size of the central metal ion. The sequence of the given ions in is $\mathrm{Mn^{2+} < Fe^{2+} < Co^{2+} < Ni^{2+} < Cu^{2+} < Zn^{2+}}$**. This sequence is known as the Irving-William order of stability of the complexes of** $\mathrm{M^{2+}}$ **ions.**

Q.12 Answer:

$_{29}\mathrm{Cu}^{+}$

Solution:

The electronic configurations are: $_{25}\mathrm{Mn^{2+}(3d)^5}; _{29}\mathrm{Cu^{+}(3d)^{10}}; _{27}\mathrm{Co^{3+}(3d)^6}; _{29}\mathrm{Cr^{3+}(3d)^3}$ **.** $_{29}\mathrm{Cu}^{+}$ **has no unpaired electrons, so it is diamagnetic.**

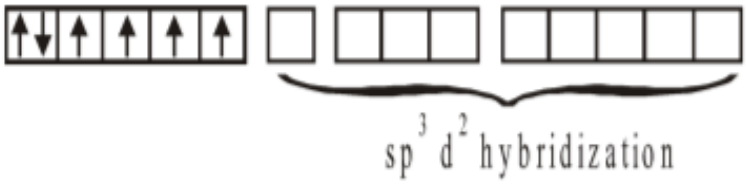
Q.13 Answer:

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Cobalt involves $sp^3\ d^2$ hybridization

Solution:

Cobalt in the complex exists as



Q.14 Answer:

$_{29}Cu$

Solution:

The electronic configuration of Cu is
$(3d)^{10}(4s)^1$

Q.15 Answer:

$CoCl_3$

Solution:

In $FeCl_2$ configuration is $3d^64s^0$ i.e.,



four unpaired electrons.
Only $CoCl_3$ has a similar configuration $3d^64s^0$ and therefore four unpaired electrons.

Q.16 Answer:

$TiCl_3$

Solution:

Only $TiCl_3$ has one unpaired electron in the lower d orbital which can be involved in $d-d$ transition $Ti\,(SO_4)_2$ and $ScCl_3$ have no electrons in $(n-1)d$ where as Cu_2Cl_2 has $(n-1)d^{10}$ configuration.

Q.17 Answer:

IO_3^-

Solution:

$$2KMnO_4 + KI + H_2O \rightarrow 2KOH + 2MnO_2 + KIO_3$$

Q.18 Answer:

$ZnCl_2$

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Solution:

Zn^{2+} **in** $ZnCl_2$ **has a** d^{10} **configuration where all the electrons are paired up and thus** $ZnCl_2$ **is diamagnetic.**

Q.19 Answer:

$TiO > VO > CrO > FeO$

Solution:

Basic strength decreases across the period.

Q.20 Answer:

d^5

Solution:

d^5 **contains** 5 **unpaired electrons, hence it shows maximum magnetic moment.**

Q.21 Answer:

Na_2ZnO_2

Solution:

$$Zn + NaOH \longrightarrow Zn(OH)_2$$
$$Zn(OH)_2 + NaOH \longrightarrow Na_2\left[Zn(OH)_4\right]$$
or $Na_2ZnO_2.2H_2O$

Q.22 Answer:

$+3$

Solution:

Most inner-transition elements show the valency of $+3$**.**

Q.23 Answer:

$Yb^{+3} < Pm^{+3} < Ce^{+3} < La^{+3}$

Solution:

Due to Lanthanoid contraction order will be $Yb^{+3} < Pm^{3+} < Ce^{+3} < La^{3+}$

Q.24 Answer:

$Ti^{2+}, V^{3+}, Cr^{4+}, Mn^{5+}$

Solution:

$Ti_{22} = 3d^2\ 4s^2; Ti^{2+} = 3d^2$
$V_{23} = 3d^3\ 4s^2; V^{3+} = 3d^2$

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$Cr_{24} = 3d^4\ 4s^2; Cr^{4+} = 3d^2$
$Mn_{25} = 3d^5\ 4s^2; Mn^{5+} = 3d^2$

Q.25 Answer:

Np

Solution:

The transuranic elements are the chemical elements with atomic numbers greater than 92 (the atomic number of Uranium). Neptunium is the first element of transuranic series with atomic number
93

Q.26 Answer:

all of these

Solution:

Because of lanthanoid contraction, Zr(40) and Hf(72) have nearly the same atomic radii.\n

Element	Radius (ppm)
Zr(40)	145
Hf(72)	144
Ta(73)	134
Ti(22)	132
Nb(41)	134

Q.27 Answer:

Pu, Am, Cm

Solution:

Pu, Am **and** Cm **are prepared from uranium, hence called uranides.**

Q.28 Answer:

Decreases

Solution:

Across the lanthanide series the size of the cations decreases thereby decreasing ionic character and increasing covalent character of hydroxides of lanthanides. Thus, the basicity of hydroxides decreases.

Q.29 Answer:

14

Solution:

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Lanthanides (total 14) are called rare earth elements. Hence, total number of rare earth elements are 14 .

Q.30 Answer:

Third

Solution:

It should be placed in the third group as the element loses $5d^1$ and $6s^2$ electrons to attain $+3$ oxidation state easily.

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